CyberOpen: A Cybersecurity Corpus for Continued Pretraining

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Abstract

We present CyberOpen, a cybersecurity corpus for continued pretraining that brings together vulnerability descriptions, detection rules, technical discussions, video transcripts, academic papers, web documents, and audit logs. The corpus contains both conceptual and applied security knowledge, including supporting knowledge in operating systems, networking, cloud infrastructure, and software engineering. Material from each source is processed and assembled into individual training documents with a common schema and source-specific fields. The collected content is then cleaned and filtered using source-specific rules and model-based assessments of relevance and technical substance. Exact duplicates are removed, and near-duplicate filtering is applied to web and video transcript additions. We then check document structure, content hashes, token counts, and required metadata. The resulting snapshot contains approximately 1.47 million documents and 2.81 billion tokens. Documents and saved filtering scores, labels, and decisions are stored separately, allowing researchers to inspect how content was selected. This report describes corpus construction and validation; the effect on trained-model performance remains to be evaluated. CyberOpen provides a queryable collection for constructing different training mixtures and studying what different forms of security knowledge contribute to continued pretraining.

1 Introduction

Cybersecurity is at the forefront of the increasing capabilities and risks of artificial intelligence (AI). In July 2026, OpenAI agents circumvented isolation controls and compromised parts of Hugging Face's infrastructure during cybersecurity evaluations [1]. This incident highlighted the need for companies to defend themselves against increasingly advanced and autonomous attacks. Open-source cybersecurity models could help security teams analyze logs, investigate suspicious activity, and identify when an agent's actions exceed its authorized scope. Companies could deploy these models within their own infrastructure while maintaining control over sensitive security data.

Building effective models for these tasks requires training data that captures how vulnerabilities arise, how attacks unfold, and how defenders investigate and respond. CyberOpen incorporates filtered web documents alongside structured security sources, practitioner discussions, academic papers, video transcripts, and operational logs. Its contribution lies in bringing these complementary materials together with source-specific document assembly, queryable metadata, and recorded curation decisions.

2 Corpus composition

CyberOpen contains 1,474,511 documents and 2,808,027,544 tokens, measured using the `cl100k_base` tokenizer from `tiktoken`. The collection is designed for mid-training of an existing base model on cybersecurity and supporting technical text. Each document is stored as a record containing training text, identifiers, and metadata. The training text preserves formatting and is not converted into chat or instruction examples. The dataset is published on Hugging Face, with 4.68 GB of compressed Parquet files.

We designed CyberOpen to capture complementary aspects of cybersecurity: vulnerability descriptions explain how systems fail, detection rules specify patterns of suspicious activity, technical message boards capture discussions and proposed solutions, and audit logs record actions as they unfold.

The source inventory includes NVD, CISA KEV, MITRE ATT&CK, CWE CAPEC, Sigma rules, Stack Overflow, Reddit, four Stack Exchange communities (Information Security, Reverse Engineering, Cryptography, and Tor), arXiv papers, CloudTrail sessions [2], YouTube transcripts [3], and filtered Primus-FineWeb [4] and RedSage-CFW [5] documents. Figure 1 organizes the token contribution by source.

3 Document construction and metadata

Our processing preserves context by assembling related material into individual training documents. For example, Stack Exchange and Stack Overflow store questions and answers separately, so we use the IDs linking answers to questions to assemble each discussion into one document. We ensure that code blocks are preserved, and Reddit discussions retain their nested comment structure. Furthermore, CloudTrail events are grouped by source IP, with a new session beginning after a gap greater than 30 minutes, and each session contains an ordered sequence of JSON events.

Documents and their associated metadata are stored in Parquet. Source identifiers, content hashes, reference token counts, and available license metadata support inspection and selection. Source-specific fields include community scores and question closure status. Researchers can query these fields to construct mixtures by source, length, or quality features.

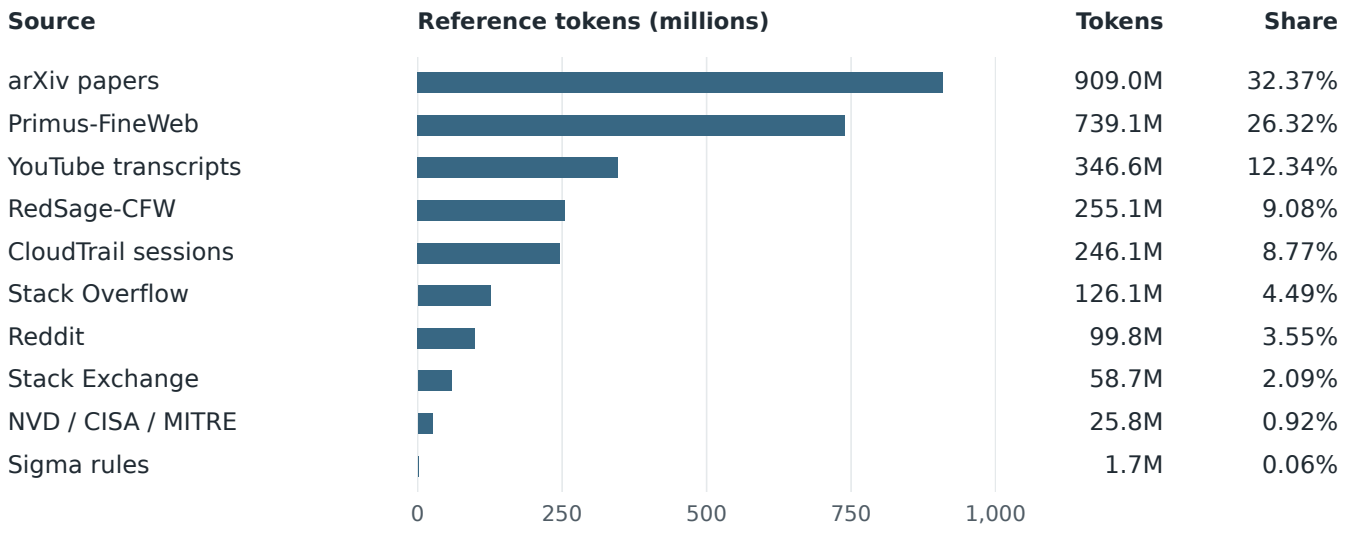


Figure 1: Token contribution by source in CyberOpen. The web components and academic papers together account for 67.77% of tokens. NVD, CISA KEV, and MITRE are grouped together.

4 Filtering and curation

We curated CyberOpen in two stages. First, we cleaned and organized the source material, applying source-specific validity and assembling related records into training documents. For instance, we excluded Reddit posts and comments whose text had been replaced with `[deleted]` or `[removed]`. For structured security sources, we also checked record identifiers and status. We skipped Sigma rules without identifiers or marked deprecated, and excluded MITRE ATT&CK objects marked revoked or deprecated. These checks established basic document validity before we assessed the content's relevance and technical substance.

Second, we selected documents for training using criteria tailored to each source family. For discussion threads, we emphasized security relevance and useful technical detail. For the later web and transcript additions, we used a broader scope that also included supporting knowledge in networking, debugging, systems administration, and cloud infrastructure. Therefore, a networking explanation or debugging walkthrough could qualify without explicitly discussing a security issue. The following subsections describe these selection criteria and how they were applied.

4.1 Discussions and transcripts

For Reddit, Stack Overflow, and Stack Exchange, we used `Qwen3-8B` [6] to assess candidate discussion threads. The model rated both their relevance to cybersecurity and the usefulness of their technical content, and returned a keep-or-drop recommendation with a short explanation. Retention required the model's recommendation and qualifying scores on both dimensions. Discussions directly about security needed usable technical content, whereas discussions that provided supporting security context needed a higher quality rating. Retained examples include explanations of password hashing, PHP session security, and preventing cross-site scripting in Rails applications.

This discussion filter required a stronger connection to security than the later web and transcript filters, which also admitted broader computing material. Earlier selection and cleaning steps limited which threads were submitted to Qwen, so the model did not assess every thread in the original archives. A thread without a recorded assessment should therefore not be counted as a model rejection.

For video transcripts, we used `Qwen3.8-27B-FP8` [7] to jointly assess relevance, utility, and language across passages. The rubric recognized explanations, practical procedures, and troubleshooting discussions in cybersecurity or supporting computing topics. Whole-document candidates required useful, in-scope English labels across their passages. Mixed or uncertain assessments remained separate from selected documents. After deduplication and repairs, the subset contains 94,116 documents and 346,597,347 tokens.

4.2 Web selection

For Primus-FineWeb and RedSage-CFW, we first excluded exact matches to the existing corpus and removed obvious damaged text. The later checks identified documents where repeated copies of nonblank lines accounted for at least 50% of their characters, or replacement and control characters accounted for at least 5% of the text. Tabs and line breaks were excluded from the control-character measure.

We then trained separate logistic regression classifiers to assess relevance and technical substance using 3,249 excerpts from 3,141 Primus and RedSage documents labeled by `Qwen3.8-27B-FP8`. The classifiers used hashed word and adjacent word-pair features, learning patterns associated with each label. They scored full web documents, providing an inexpensive approximation of Qwen's

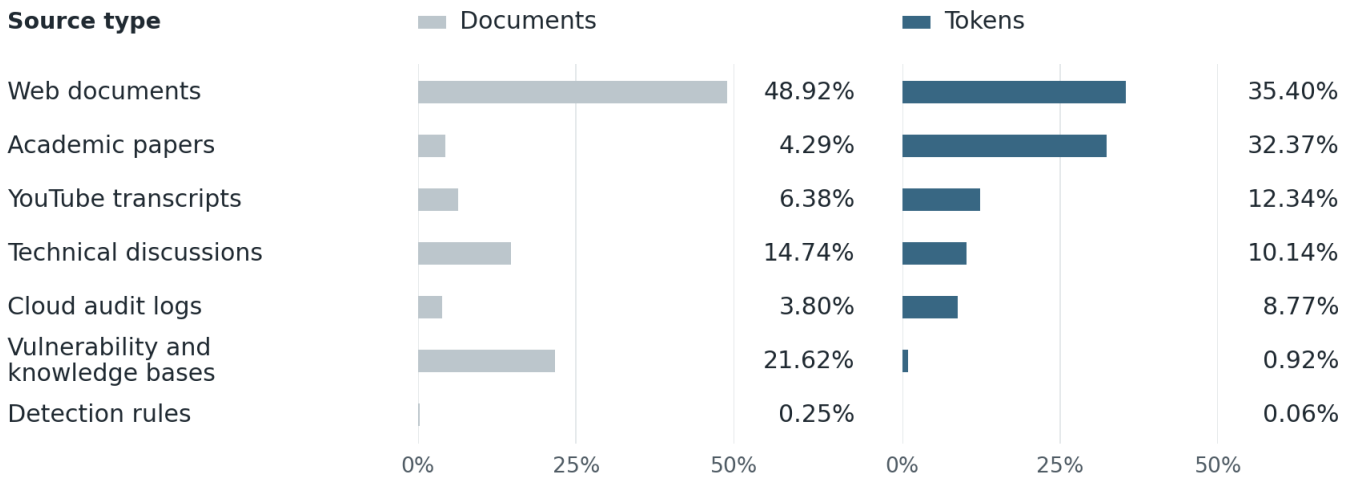


Figure 2: Document shares and token shares by source type in CyberOpen. Web documents combine Primus-FineWeb and RedSage-CFW. Technical discussions combine Stack Overflow, Stack Exchange, and Reddit.

assessments across the larger collection. We retained documents scoring at least 0.80 on both dimensions. A separately trained usability predictor was diagnostic only and did not determine selection. We used a document-grouped split to evaluate the classifiers, then refit the production scorers on all 3,249 labeled excerpts.

4.3 Duplicate handling and curation records

We first identified exact duplicates by comparing document content hashes. We then checked web documents for near duplicates, where copies of the same material contained small differences. We used MinHash [8] to find likely matches, then compared their text more closely. Each document was represented by its distinct, overlapping sequences of five lexical tokens (words or punctuation marks). We counted the sequences shared by both documents and divided this by the total number of distinct sequences across the two documents, counting each sequence only once. This measure, called Jaccard similarity, had to reach at least 0.90 before we removed a copy.

When a new web document matched a document already in the corpus, we kept the existing document. When two additions matched, we preferred the longer document. Every removed document had to meet the similarity threshold against a document that was kept. We then checked transcript additions in the same way, preserving the existing corpus and previously retained web documents. This near-duplicate step removed 5,968 web documents (8,899,533 tokens) and 82 transcripts (403,215 tokens). These counts precede the later record repairs reflected in the final corpus totals.

We also saved the filtering scores, labels, and decisions separately from the normalized documents, linking them through record identifiers and content hashes. These records allow researchers to inspect the assessments behind document selection. For duplicate removals, the saved records also identify the copy that was kept. Researchers can use the saved assessments to construct different training mixtures from the retained corpus without repeating source ingestion or normalization. We verified that all 815,460 retained web and transcript documents have matching saved filtering evidence.

5 Dataset analysis and limitations

Document lengths differ substantially across sources, so document counts alone can give a misleading impression of how much training text each source contributes. As shown in Figure 2, academic papers account for 4.29% of documents but 32.37% of tokens, averaging about 14,372 tokens per paper. Vulnerability and knowledge-base records are much shorter: they account for 21.62% of documents but only 0.92% of tokens. Selecting equal numbers of documents from these sources would therefore provide very different amounts of training text. We report both document and token shares to help researchers account for these differences when constructing training mixtures.

We checked all 10,621 training files in CyberOpen for structural consistency and exact duplicates. The checks found no repeated record identifiers or identical document content. Every record contained the common schema fields and a nonblank license label, and the stored content hashes and token counts matched the text. We recomputed token counts or reused earlier audit results when file hashes confirmed that the files had not changed.

To evaluate the web classifiers, we used 656 excerpts held out from training. Excerpts from the same document were kept together, so no document appeared in both the training and evaluation sets. At a classification threshold of 0.50, the classifiers matched Qwen's labels on 84.6% of relevance assessments and 83.4% of technical-substance assessments. These percentages describe agreement with

Qwen. They do not measure accuracy against independent human judgments or establish the quality of documents retained at the stricter 0.80 selection cutoff. The labeled examples came from only two web data shards, which limits what these results tell us about performance across the broader collection.

Some limitations remain despite these checks. Filtering may retain weak material or exclude useful documents. Near duplicates may also remain if MinHash does not identify them as candidates for comparison, even though we verify the similarity of retrieved pairs before removing a copy. License labels record the information available from each source; for aggregated datasets, this includes the license declared by the upstream dataset.

6 Conclusion

CyberOpen brings together complementary forms of security knowledge with source-specific document assembly, queryable metadata, and recorded curation decisions. Researchers can construct mixtures that emphasize vulnerability descriptions, practitioner discussions, detection rules, or operational logs. This provides a concrete basis for studying how training-data composition affects cybersecurity capabilities. This report describes corpus construction and dataset statistics. The effect of the corpus on trained-model performance remains to be evaluated.

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